9/10/26 1) Genetic Optimization of algorithm.

Algorithm:-

step 1: Select a population
step 2: use the cromosomes of the population
step 3: evaulate cromozome fitness using fitness function
step 4: The value of fitness are stored, select the optimal cromolomes
step 5: if optimal value matches terminal condition stop the pross
step 6: select the optimal values and crossover them
step 7 :- perform mutation for divergence in offspring
step 8: Take optimal population from offspring possible ~~Parents also~~ parent generation
step 9: repeat step 1 to 8
~~step 9:~~
step 10: use the result to produce offspring

Renewable Energy Optimization

Optimize the operation of solar wind and Battery system minimize energy cost and improve renewable energy utilization

BEGIN

input solar_energy, wind_energy, energy_demand, battery_capacity, electricity_cost.

Initialize population with random solutions.

For each generation DO

For each solution in population DO

CALculate renewable_energy

calculate battry_operation

calculate total_energy_required

calculate renewable_energy_utilization

Calculate fitness based on minimum cost and max renewable utilization

End For

Select best solution

Perform crossover

Perform mutation

Check energy demand and battery constraints

Update population with new sol^n

End For

Select best feasible

Problem Statement: solar and wind energy availability varies with weather conditions, making efficient energy management challenging. Improper utilization leads to energy wastage and increased electricity cost. The project uses Genetic Algorithm to optimize solar, wind and battery operations.

Objectives: To minimize total energy cost maximize renewable energy utilization, and meet energy demand while maintaining battery constraints.

Specific O/P Seen.

Output:

Optimal Battery Schedule (kw per hour):

[-4.85 -13.82 -8.92 17.59 4.49 -4.67 -5.82
-12.51 16.45 9.11 8.48 17.6 18.71 9.4 -0.62
4.14 -24.14 -19.98 -8.63 7.7 6.13 1.92
-19.34 -9.14]

Best Fitness Value: 0.025585565634362426